Principal Component Analysis – Algorithm

Step 1. Data

We consider a dataset having $n$features or variables denoted by $X_1, X_2, \ldots, X_n$.
Let there be Nexamples.
Let the values of the i^{th}feature $X_i$be $X_{i1}, X_{i2}, \ldots, X_{iN}$.

Features	**Example 1**	**Example 2**	**…**	**Example N**
X_1	X_{11}	X_{12}	...	X_{1N}
X_2	X_{21}	X_{22}	...	X_{2N}
$\vdots$	$\vdots$	$\vdots$	...	$\vdots$
X_i	X_{i1}	X_{i2}	...	X_{iN}
$\vdots$	$\vdots$	$\vdots$	...	$\vdots$
X_n	X_{n1}	X_{n2}	...	X_{nN}

Step 2. Compute the means of the variables

$$\bar{X}_i = \frac{1}{N}(X_{i1} + X_{i2} + \cdots + X_{iN})$$

Step 3. Calculate the covariance matrix

$$S = \begin{bmatrix} \mathrm{Cov}(X_1, X_1) & \mathrm{Cov}(X_1, X_2) & \ldots & \mathrm{Cov}(X_1, X_n) \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Cov}(X_2, X_2) & \ldots & \mathrm{Cov}(X_2, X_n) \\ \vdots & \vdots & \ldots & \vdots \\ \mathrm{Cov}(X_n, X_1) & \mathrm{Cov}(X_n, X_2) & \ldots & \mathrm{Cov}(X_n, X_n) \end{bmatrix}$$

$$\mathrm{Cov}(X_i, X_j) = \frac{1}{N-1}\sum_{k=1}^{N}(X_{ik} - \bar{X}_i)(X_{jk} - \bar{X}_j)$$

Step 4. Calculate the eigenvalues and eigenvectors of the covariance matrix

i. Set up the equation. This is a polynomial equation of degree n in λ. It has n real roots and these roots are the eigenvalues of S.

$$\det(S - \lambda I) = 0$$

ii. If $\lambda = \lambda'$is an eigenvalue, then the corresponding eigenvector is a vector

$$U = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \text{ such that} (S - \lambda' I)U = 0$$

iii. We now normalize the eigenvectors. Given any vector X, we normalize it by dividing $X$by its length. The length (or norm) of the vector

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \text{ is defined as } \parallel X \parallel = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

We compute the $n$normalized eigenvectors $e_1, e_2, \ldots, e_n$by

$$e_i = \frac{1}{\parallel U_i \parallel} U_i, i = 1,2, \ldots, n$$

Step 5. Derive new dataset

Order the eigenvalues from highest to lowest.
The unit eigenvector corresponding to the largest eigenvalue is the first principal component.

i. Let the eigenvalues in descending order be

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$$

and let the corresponding unit eigenvectors be $e_1, e_2, \ldots, e_n$.

ii. Choose a positive integer p such that $1 \leq p \leq n$.

iii. Choose the eigenvectors corresponding to the eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_p$and form the following $p \times n$matrix (we write the eigenvectors as row vectors):

$$F = \begin{bmatrix} e_1^T \\ e_2^T \\ \vdots \\ e_p^T \end{bmatrix}$$

iv. We form the following $n \times N$matrix:

$$X = \begin{bmatrix} X_{11} - \bar{X}_1 & X_{12} - \bar{X}_1 & \ldots & X_{1N} - \bar{X}_1 \\ X_{21} - \bar{X}_2 & X_{22} - \bar{X}_2 & \ldots & X_{2N} - \bar{X}_2 \\ \vdots & \vdots & \ldots & \vdots \\ X_{n1} - \bar{X}_n & X_{n2} - \bar{X}_n & \ldots & X_{nN} - \bar{X}_n \end{bmatrix}$$

v. Next, compute the matrix-

$$X_{new} = FX$$

Note that this is a $p \times N$ matrix. This gives us a dataset of N samples having p features.

Principal Component Analysis –Example

Given the data in the table, reduce the dimension from **2 to 1** using the **Principal Component Analysis (PCA)** algorithm.

Feature	Example 1	Example 2	Example 3	Example 4
X_1	4	8	13	7
X_2	11	4	5	14

Step 1: Calculate Mean

$$\bar{X}_1 = \frac{1}{4}(4 + 8 + 13 + 7) = 8$$
$$\bar{X}_2 = \frac{1}{4}(11 + 4 + 5 + 14) = 8.5$$

Step 2: Calculation of the Covariance Matrix

$$S = \begin{bmatrix} \mathrm{Cov}(X_1, X_1) & \mathrm{Cov}(X_1, X_2) \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Cov}(X_2, X_2) \end{bmatrix}$$

$$\mathrm{Cov}(X_1, X_1) = \frac{1}{N-1}\sum_{k=1}^{N}(X_{1k} - \bar{X}_1)(X_{1k} - \bar{X}_1)$$
$$= \frac{1}{3}((4-8)^2 + (8-8)^2 + (13-8)^2 + (7-8)^2)$$
$$= 14$$

$$\mathrm{Cov}(X_1, X_2) = \frac{1}{N-1}\sum_{k=1}^{N}(X_{1k} - \bar{X}_1)(X_{2k} - \bar{X}_2)$$
$$= \frac{1}{3}((4-8)(11-8.5) + (8-8)(4-8.5) + (13-8)(5-8.5) + (7-8)(14-8.5))$$
$$= -11$$

$$\mathrm{Cov}(X_2, X_1) = \mathrm{Cov}(X_1, X_2) = -11$$

$$\mathrm{Cov}(X_2, X_2) = \frac{1}{N-1}\sum_{k=1}^{N}(X_{2k} - \bar{X}_2)(X_{2k} - \bar{X}_2)$$

$$= \frac{1}{3}((11 - 8.5)^2 + (4 - 8.5)^2 + (5 - 8.5)^2 + (14 - 8.5)^2)$$

$$= 23$$

Covariance Matrix S:

$$S = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

Step 3: Eigenvalues of the Covariance Matrix

The characteristic equation of the covariance matrix is

$$\begin{aligned}
0 &= \det(S - \lambda I) \\
&= \begin{vmatrix} 14 - \lambda & -11 \\ -11 & 23 - \lambda \end{vmatrix} \\
&= (14 - \lambda)(23 - \lambda) - (-11)(-11) \\
&= \lambda^2 - 37\lambda + 201 \\
\lambda &= \frac{1}{2}(37 \pm \sqrt{565}) \\
&= 30.3849,\ 6.6151 \\
&= \lambda_1,\ \lambda_2 \ \text{(say)}
\end{aligned}$$

Step 4: Computation of the Eigenvectors

$$U = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = (S - \lambda I)U$$

$$= \begin{bmatrix} 14 - \lambda & -11 \\ -11 & 23 - \lambda \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$= \begin{bmatrix} (14 - \lambda)u_1 - 11u_2 \\ -11u_1 + (23 - \lambda)u_2 \end{bmatrix}$$

$$(14 - \lambda)u_1 - 11u_2 = 0$$

$$-11u_1 + (23 - \lambda)u_2 = 0$$

Both equations represent the **same line**, so you only need **one equation**.

Use the first equation

$$(14 - \lambda)u_1 - 11u_2 = 0$$

Move one term:

$$(14-\lambda)u_1 = 11u_2$$

Divide both sides:

$$\frac{u_1}{11} = \frac{u_2}{14-\lambda}$$

Now set them equal to a parameter t:

$$\frac{u_1}{11} = \frac{u_2}{14-\lambda} = t$$

Multiply back:

$$u_1 = 11t$$
$$u_2 = (14-\lambda)t$$

Factor out t:

$$U = t\begin{bmatrix} 11 \\ 14-\lambda \end{bmatrix}$$

Since eigenvectors ignore scalar multiples, take $t = 1$:

$$U = \begin{bmatrix} 11 \\ 14-\lambda \end{bmatrix}$$

From

$$(14-\lambda)u_1 - 11u_2 = 0$$

Immediately write

$$u_1 : u_2 = 11 : (14-\lambda)$$

So eigenvector

$$U = \begin{bmatrix} 11 \\ 14-\lambda \end{bmatrix}$$

Given

$$\lambda_1 = 30.3849$$

So

$$14 - \lambda_1 = 14 - 30.3849 = -16.3849$$

Thus, the eigenvector becomes

$$U_1 = \begin{bmatrix} 11 \\ -16.3849 \end{bmatrix}$$

Find the length (norm)

Formula for vector length:

$$\| U \| = \sqrt{x^2 + y^2}$$

So

$$\| U_1 \| = \sqrt{11^2 + (-16.3849)^2}$$

Calculate:

$$\begin{aligned} &11^2 = 121 \\ &(-16.3849)^2 = 268.75 \\ &121 + 268.75 = 389.75 \\ &\sqrt{389.75} = 19.7348 \end{aligned}$$

So

$$\| U_1 \| = 19.7348$$

Make it a unit vector

Divide each component by the length.

$$e_1 = \begin{bmatrix} 11/19.7348 \\ -16.3849/19.7348 \end{bmatrix}$$

Final result

$$11/19.7348 = 0.5574$$
$$-16.3849/19.7348 = -0.8303$$

So

$$e_1 = \begin{bmatrix} 0.5574 \\ -0.8303 \end{bmatrix}$$

Second eigenvector

Similarly

$$e_2 = \begin{bmatrix} 0.8303 \\ 0.5574 \end{bmatrix}$$

Step 5: Computation of first principal components

We calculate

$$e_1^T \begin{bmatrix} X_{1k} - \bar{X}_1 \\ X_{2k} - \bar{X}_2 \end{bmatrix}$$

Unit eigenvector × centered data point

First eigenvector

From previous step

$$e_1 = \begin{bmatrix} 0.5574 \\ -0.8303 \end{bmatrix}$$

Transpose means row vector

$$e_1^T = [0.5574 \quad -0.8303]$$

First observation

$$X_{11} = 4$$
$$X_{21} = 11$$

Means

$$\bar{X}_1 = 8$$
$$\bar{X}_2 = 8.5$$

So centered vector

$$\begin{bmatrix} 4 - 8 \\ 11 - 8.5 \end{bmatrix} = \begin{bmatrix} -4 \\ 2.5 \end{bmatrix}$$

Multiply vectors

Now multiply

$$[0.5574 \quad - 0.8303] \begin{bmatrix} -4 \\ 2.5 \end{bmatrix}$$

Multiply term by term

$$= 0.5574(-4) + (-0.8303)(2.5)$$

First part

$$0.5574 \times (-4) = -2.2296$$

Second part

$$-0.8303 \times 2.5 = -2.07575$$

Add

$$-2.2296 - 2.07575$$
$$= -4.30535$$

Final result

$$-4.30535$$

Feature	Ex 1	Ex 2	Ex 3	Ex 4
X_1	4	8	13	7
X_2	11	4	5	14
First Principal Components	-4.3052	3.7361	5.6928	-5.1238

Step 6: Geometrical meaning of first principal components